DIGITAL LOGIC DESIGN (LAB)

PROJECT PROPOSAL

Project Name: ***RotorScope***

Group #: *13*



Program: ***BS-CS***

Semester: ***Spring 2024***

Faculty Name: ***Ms. Mahrose Fatima Naqvi***

S.no	Reg No	Name	
1	25949	Mohammad Bin Farooq	**Group Leader**
2	19288	Mohammad Omer Shaikh	**Group Member**
3	21296	Kabeer Hussain	**Group Member**

MICROCONTROLLER BASED TACHOMETER

Objective:

The objective for the tachometer project could be to design and develop a microcontroller based system that accurately measures and displays the rotational speed of an engine or any mechanical device. It's all about precision and real-time monitoring.

Motivation:

Our motivation could be to create a cost effective, efficient, and reliable tool that enhances the diagnostic capabilities for machinery. It could also be about providing a learning platform for understanding embedded systems and their applications in the real world. This project can serve as a practical example of applying microcontroller technology to solve engineering problems. Also it is a great way to put your skills to the test and create something useful.

Project's Target Audience:

- People who fix motors and engine which needed to check engine speeds.
- Engineers who want to keep an eye on machine parts.
- Anyone who likes to build and create things for fun.
- Could be useful in DIY.

List of Key Features:

- Accurate RPM measurement: It should give precise readings of accurate speed.
- Real time display: The RPM data should be shown instantly on a screen.
- User friendly interface: Easy for anyone to use with the clear instructions.
- Durable and portable design: It can be used in different environments and easy to carry around.

Innovation (Apart from Project's Typical Objective):

Innovate by adding a smart alert that notifies you when there's an unusual reading.

Signature:
Name: Mohammad Bin Farooq
Group Leader

Signature:
Name: Mohammad Omer Shaikh
Group Member

Signature:
Name: Kabeer Hussain
Group Member